



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

31-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,D	A,B	B,C	B,C	A,D	A		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	A	A	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	121.69 to 121.70	120.00	4.80	0.50	16.00	0.40	0.13	100.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,D	A,D	A,C,D	A,B,D	A,B,D	A,B,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	C	D	A	D				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	3.00	7.00	0.82	21.48	209.00	20.00	6.00	2.40

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	C,D	B,C	A,B,C,D	A,D	A,C	A,C,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	A	B	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	0.40 to 0.43	0.10	5.00	1.72 to 1.74	1.50	1.57	0.00	0.29 or 0.30

HINT – SHEET

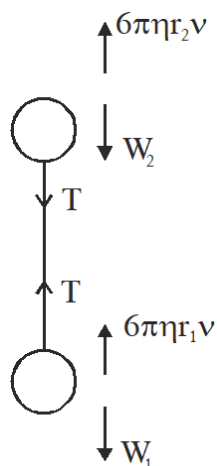
PART-1 : PHYSICS

SECTION-I (i)

1. Ans (A,D)

$$\rho \left[\frac{4}{3} \pi r_1^3 \right] g + \rho \left[\frac{4}{3} \pi r_2^3 g \right]$$

$$= 6\pi\eta r_1 v + 6\pi\eta r_2 v$$



$$v_t = \frac{2\rho g}{9\eta} [r_1^2 - r_1 r_2 + r_2^2]$$

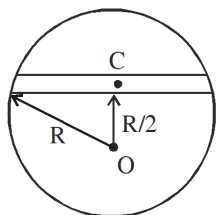
$$\therefore \frac{4}{3} \pi r_1^3 \rho g = T + 6\pi\eta r_1 v$$

put v and find T.

2. Ans (A,B)

$$Wd = \Delta U = -\frac{GMm}{2R^3} \left[3R^2 - \frac{R^2}{4} \right] - \frac{GMm}{R}$$

$$= -\frac{3GMm}{8R}$$



also $N = F \cos \theta$ (at any position θ)

$$\frac{GMm}{R^3} \left(\frac{R}{2} \cos \theta \right) = N$$

$$\mu N = \frac{3GMm}{8R}$$

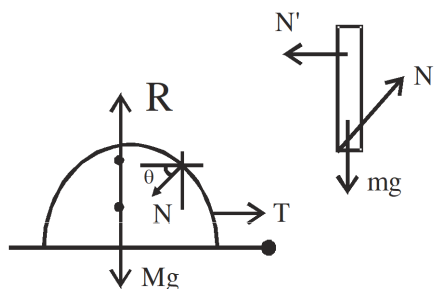
$$\mu = \frac{\sqrt{3}}{2}$$

3. Ans (B,C)

$$R = Mg + N \sin \theta$$

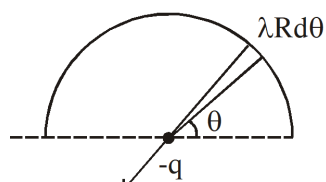
$$T = N \cos \theta$$

$$N \sin \theta = mg$$



4. Ans (B,C)

$$U = \frac{K(q)(-q)}{R} = -\frac{Kq^2}{R}$$



$$dE = \frac{K\lambda R d\theta}{R^2}$$

$$dE_{\text{net}} = \frac{2K\lambda}{R} \sin \theta d\theta$$

$$F = q \frac{2K\lambda}{R} \int_0^{\pi/2} \sin \theta d\theta$$

$$F = \frac{2Kq^2}{\pi R^2}$$

5. Ans (A,D)

$$\delta_m = 60^\circ, i = 60^\circ$$

$$\therefore i = e = 60^\circ$$

$$\delta_m = 2i - A \Rightarrow 60^\circ = 2(60^\circ) - A \Rightarrow A = 60^\circ$$

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2} = \sqrt{3}$$

$$\delta = i_1 + 2i_1 - A \Rightarrow i_1 = 41^\circ$$

6. Ans (A)

$$\text{At 'P' } u_x = 20 \sin 37^\circ = 12 \text{ ms}^{-1}$$

$$u_y = 16 \text{ ms}^{-1}$$

$$\text{at 'Q' } u_x = 12 \text{ ms}^{-1}$$

$$\frac{V_x}{V_y} = \tan 53^\circ \Rightarrow V_y = 9 \text{ ms}^{-1}$$

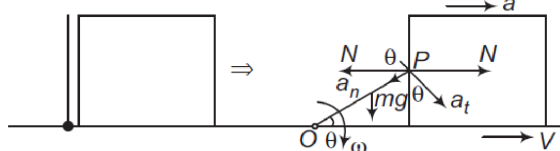
$$\therefore -9 = 16 - 10t \Rightarrow t = 2.5 \text{ s}$$

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (C)**

At angle θ shown in figure :



Decrease in potential energy of rod = increase in rotational kinetic energy of rod + translational kinetic energy of block.

But $v = v_P \sin \theta = (\ell \omega) \sin \theta$

$$\begin{aligned} \therefore mg \left(\frac{\ell}{2} - \frac{\ell}{2} \sin \theta \right) &= \frac{1}{2} \left(\frac{m\ell^2}{3} \omega^2 \right) + \frac{1}{2} M (\omega \ell \sin \theta)^2 \end{aligned}$$

From here we get,

$$\ell \omega^2 = a_n = \frac{mg(1 - \sin \theta)}{m/3 + M \sin^2 \theta} \quad \dots (i)$$

$$\begin{aligned} a_t = \ell \alpha &= \ell \left(\frac{\tau}{I} \right)_0 = \ell \left[\frac{mg \frac{\ell}{2} \cos \theta - N \ell \sin \theta}{m\ell^2/3} \right] \\ &= \frac{3}{2} g \cos \theta - \frac{3N \sin \theta}{m} \quad \dots (ii) \end{aligned}$$

For block

$$N = Ma = M[a_t \sin \theta - a_n \cos \theta] \quad \dots (iii)$$

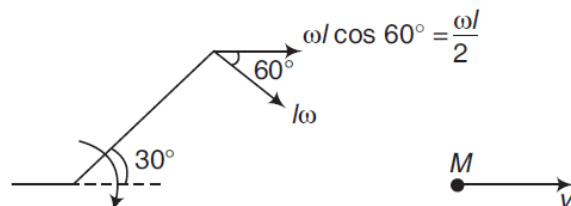
Now putting values of a_t and a_n from Eqs. (i) and (ii) in Eqs. (iii) and then putting $N = 0$ and

$\theta = 30^\circ$ in the equation we get,

$$\frac{M}{m} = \frac{4}{3}$$

8. **Ans (A)**

$$\frac{M}{m} = \frac{4}{3} \quad M = \frac{4}{3} m$$



$$\frac{\omega \ell}{2} = v \quad \therefore \omega = \frac{2v}{\ell}$$

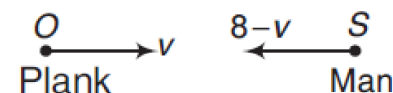
Decrease in potential energy of rod = increase in rotational kinetic energy of rod + translational kinetic energy of block

$$\begin{aligned} \therefore mg \left(\frac{\ell}{2} - \frac{\ell}{2} \sin 30^\circ \right) &= \frac{1}{2} \left(\frac{m\ell^2}{3} \right) \left(\frac{2v}{\ell} \right)^2 + \frac{1}{2} \left(\frac{4m}{3} \right) (v^2) \end{aligned}$$

Solving this equation, we get

$$v = \frac{\sqrt{3g\ell}}{4}$$

9. **Ans (A)**



From conservation of linear momentum,

$$50(8 - v) = 150v$$

$$\therefore v = 2 \text{ m/s}$$

$$8 - v = 6 \text{ m/s}$$

$$\begin{aligned} f_1 &= f_0 \left(\frac{v + v_0}{v - v_s} \right) \\ &= f_0 \left(\frac{330 + 2}{330 - 6} \right) = \frac{332}{324} f_0 \end{aligned}$$

10. **Ans (C)**



$$\begin{aligned} f_2 &= f_0 \left(\frac{v - v_0}{v + v_s} \right) \\ &= f_0 \left(\frac{330 - 2}{330 + 6} \right) = \left(\frac{328}{336} \right) f_0 \end{aligned}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (121.69 to 121.70)

Energy conservation gives

$$\text{Energy of H atom in } n^{\text{th}} \text{ state} + \frac{hc}{\lambda_1} = 12.2 \text{ eV}$$

$$\Rightarrow -\frac{13.6}{n^2} \text{ eV} + \frac{hc}{\lambda_1} = 12.2 \text{ eV}$$

$$\Rightarrow -\frac{13.6}{n^2} + \frac{6.62 \times 10^{-34} \times 10^8}{796 \times 10^{-10} \times 1.6 \times 10^{-19}} = 12.2$$

$$\Rightarrow \frac{13.6}{n^2} = 15.6 - 12.2 = 3.4 \text{ eV}$$

$$\therefore n = 2$$

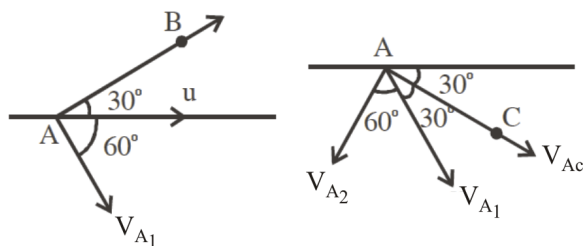
Hydrogen atom formed is in 1st excited state.

$$\therefore \frac{hc}{\lambda_2} = 10.2 \text{ eV}$$

$$\lambda_2 = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{10.2 \times 1.6 \times 10^{-19}}$$

2. Ans (120.00)

In an oblique elastic collision of two identical balls, the two will move perpendicular to each other after collision. The figure shows the centres of the balls. After first collision, B moves along the line of impact and A moves perpendicular to that. Then A hits C with velocity v_{A1} which makes 30° angle with line of impact. C goes along the impact and A is deflected perpendicular to that. v_{A2} makes 120° with original direction of motion.

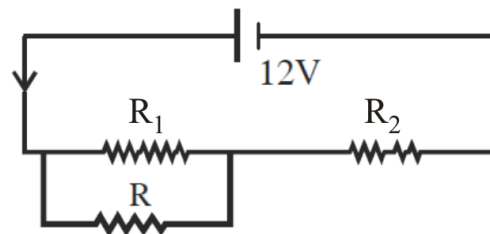


3. Ans (4.80)

The reading of voltmeter is absolutely correct when it is connected across the battery.

It means the emf of the cell is = 12 volts

Let the resistance of the voltmeter be R.



The current I is

$$I = \frac{12}{\frac{R_1 R}{R_1 + R} + R_2}$$

$\therefore$ Voltmeter reading = voltage drop across R_1 (or R)

$$\Rightarrow 4 = \frac{12}{\frac{R_1 R}{R_1 + R} + R_2} \cdot \frac{R_1 R}{R_1 + R}$$

$$\therefore 1 + \frac{R_2(R_1 + R)}{R_1 R} = 3$$

$$\Rightarrow \frac{R_2(R_1 + R)}{R_1 R} = 2$$

$$\Rightarrow \frac{R_2}{R} + \frac{R_2}{R_1} = 2 \quad \dots(1)$$

Similarly,

$$V_2 = \frac{12}{\frac{R_2 R}{R_2 + R} + R_1} \cdot \frac{R_2 R}{R_2 + R}$$

$$\Rightarrow 6 = \frac{12}{1 + \frac{R_1(R_2 + R)}{R_2 R}}$$

$$\Rightarrow \frac{R_1 R_2 + R_1 R}{R_2 R} = 1$$

$$\Rightarrow \frac{R_1}{R} + \frac{R_1}{R_2} = 1 \quad \dots(2)$$

Eliminating R between (1) and (2)

$$\frac{R_1}{R_2} = \frac{2}{3}$$

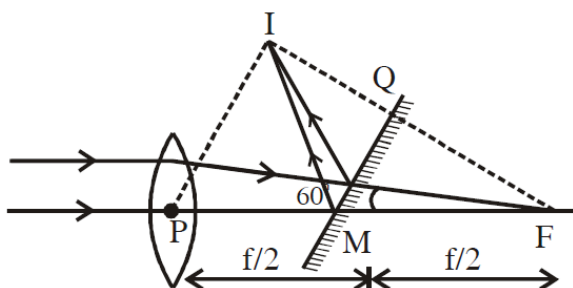
The actual voltage across R_1 is

$$V_1 = \frac{12 \cdot R_1}{R_1 + R_2} = \frac{12}{1 + \frac{R_2}{R_1}} = \frac{12}{1 + \frac{3}{2}} = \frac{24}{5} = 4.8 \text{ Volt}$$

4. Ans (0.50)

Focus point F acts as object for the plane mirror. Its image is formed at I.

From geometry, $\angle IMG = 60^\circ$ and $MF = MI = f/2$



In $\triangle PIM$ $IM = f/2$

$PM = f/2$

$\therefore IM = PM$

and $\angle PMI = 60^\circ$

Hence, $\triangle PMI$ is equilateral.

$\therefore PI = f/2$

5. Ans (16.00)

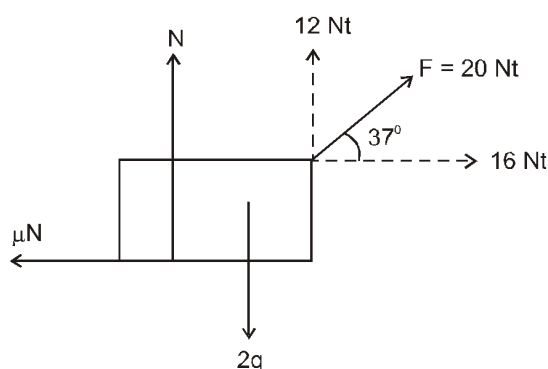
$$\mu N = 2, N = 8, a = \frac{16 - 2}{2} = 7 \text{ (}\rightarrow\text{)}$$

$$v^2 = 2(7)8 \text{ (}\rightarrow\text{)} \quad (i)$$

when the direction of horizontal component of the force F is reversed

$$a_1 = 9 \text{ m/s}^2 \text{ (}\leftarrow\text{)}$$

and distance covered by the block before it stops = $s_1 = \frac{16 \times 7}{2 \times 9}$



$$\text{Again } s_2 = 8 + s_1 = 8 + \frac{16 \times 7}{2 \times 9}$$

$$\text{and } a_2 = 7 \text{ (}\leftarrow\text{)}$$

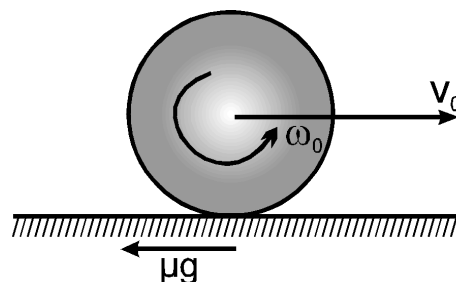
$$v^2 = 0 + 2(a_2)(s_2)$$

$$= 2(7) \left(8 + \frac{16 \times 7}{2 \times 9} \right)$$

$$\Rightarrow v = \frac{16\sqrt{7}}{3} \text{ m/s}$$

6. Ans (0.40)

Ball will come back to the initial position if its angular velocity is greater than zero in the same direction (in which it was released) at the moment its linear velocity becomes zero. In this condition ball would return back

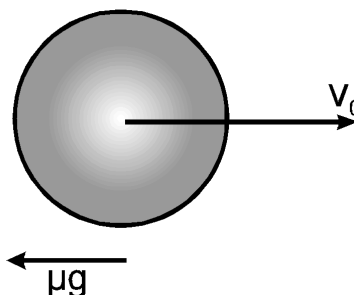


For linear motion

$$0 = v_0 - (\mu g) t$$

$$t = \frac{v_0}{\mu g} \text{ (time when ball stops)}$$

For rotation motion



$$J = I\alpha$$

$$\Rightarrow \alpha = \frac{J}{I} = \frac{\mu mg \times R}{\frac{2}{5}MR^2} = \frac{5\mu g}{2R}$$

$$\text{using } \omega_f = \omega_0 - \alpha t$$

$$\omega_f > 0$$

$$\Rightarrow \omega_0 > \alpha t$$

$$\Rightarrow \frac{v_0}{\lambda R} > \frac{5\mu g}{2R} \cdot \frac{v_0}{\mu g} \text{ for limiting condition.}$$

$$\Rightarrow \lambda_0 = \frac{2}{5}$$

PART-2 : CHEMISTRY

SECTION-I (i)

7. Ans (0.13)

$$f = \frac{v}{2(\ell + 2e)} \text{ where } e = \text{end correction} = 0.6r$$

$$\therefore f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)}$$

$$\therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{\ell + 1.2r} = \frac{\Delta v}{v} - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r}$$

$$\text{Here } \frac{\Delta v}{v} = 0 \text{ (given)}$$

$$\frac{\Delta f}{f} \times 100 = - \frac{\Delta\ell + 1.2\Delta r}{\ell + 1.2r} \times 100$$

For maximum % error : $\Delta\ell = 0.1$, $\Delta r = 0.05$

$$\left(\frac{\Delta f}{f} \times 100 \right)_{\max} = \frac{0.1 + 1.2 \times 0.05}{94 + 1.2 \times 5} \times 100 = 0.16\%$$

8. Ans (100.00)

Applying $v = u + at$ in y direction

$$100 \cos 30^\circ \tan 60^\circ = 100 \sin 30^\circ + \frac{eE}{m} t$$

{ $\because$ at entry point & exit point $v_x = 100 \cos 30^\circ$ }

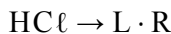
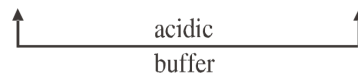
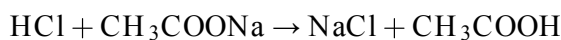
$$\therefore \text{at exit} = \frac{v_y}{100 \cos 30^\circ} = \tan 60^\circ$$

$$\Rightarrow v_y = 100 \cos 30^\circ \times \tan 60^\circ$$

$$\therefore \frac{eE}{m} t = 100 \times \frac{\sqrt{3}}{2} \cdot \sqrt{3} - 100 \times \frac{1}{2} = 100$$

$$\Rightarrow \frac{e}{m} E \cdot \frac{m}{e} = 100$$

2. Ans (A,D)



m mol of HCl react = m mol of CH_3COOH from

$$\text{pH} = \text{pK}_a + \log_b \left\{ \frac{(\text{salt})}{(\text{WA})} \right\}$$

$$4.7 = 4.7 + \log \left(\frac{\text{salt}}{\text{WA}} \right)$$

$$0 = \log_{10} \left(\frac{(\text{salt})}{(\text{WA})} \right)$$

$$\frac{[\text{Salt}]}{[\text{WA}]} = 10^0 = 1$$

$\Downarrow$

$$[\text{WA}] = [\text{Salt}] = 0.5M' [\text{CH}_3\text{COO}^\ominus]_{\text{mix}}$$

$$= [\text{CH}_3\text{COONa}] = 0.1$$

$$[\text{CH}_3\text{COOH}] = 0.5$$

$$v_f = 100\text{ml}$$

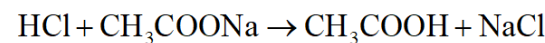
$$\text{m mol of } \text{CH}_3\text{COOH} = 0.5 \times 100$$

$$= 50 \text{ m mol} = \text{m mol of HCl. react}$$

$$M = \frac{n}{v} \Rightarrow v = \frac{n}{M} = \frac{50}{2} = 25 \text{ ml of HCl}$$

mmol of $\text{CH}_3\text{COO}^\ominus$ present in solution

$$= 0.5 \times 100 = 50 \text{ m mol}$$



$$t = 0 : 50 \text{ m mol } 50 + 50 \quad 0 \quad 0$$

$$= 100 \text{ m mol}$$

$$t_{\text{comp.}} : 0 \quad 100-50 \quad 50 \text{ m mol } 50 \text{ m mol}$$

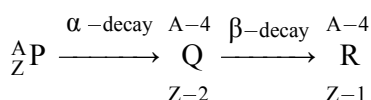
Total m mol of CH_3COONa taken initial

$$= 100 \text{ m mol}$$

$$= 100 \times 10^{-3} \times 82$$

$$= 8.2 \text{ gm}$$

3. Ans (A,C,D)



after 2 hours (i.e. 120 minute)

$$T = n \times t_{\frac{1}{2}}$$

$$120 = n \times 30$$

$$n = 4$$

$$N_t = N_0 \left(\frac{1}{2}\right)^n = N_0 \left(\frac{1}{2}\right)^4$$

$$= \frac{N_0}{16}$$

$$< 10\%$$

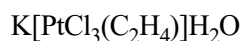
Amount of Q at any time before 30 minutes is

less than 50% as Q disintegrate to give R.

4. Ans (A,B,D)

The formula of Zeise's salt is $K[PtCl_3(C_2H_4)] \cdot H_2O$.

Let the oxidation state of Pt is x.



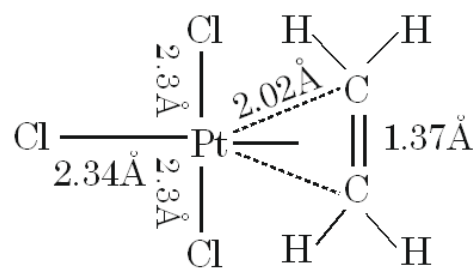
$$+1 + x + (-1) \times 3 + 0 + 0 = 0$$

$$+1 + x - 3 = 0$$

$$x = +2$$

$$Pt = +2$$

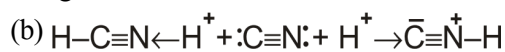
The structure of Zeise's salt is



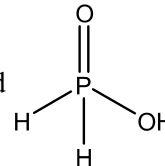
(Normal C=C length = 1.34 Å)

5. Ans (A,B,D)

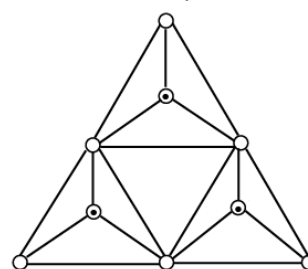
(a) Sodium pentaphosphate is $Na_7P_5O_{16}$. Thus, pentaphosphate anion is, i.e., this anion carries -7 charge.



(c) Hypophosphorous acid



(d) The primary unit of $BaTi(SiO_9)$ or $(Si_3O_9)^{6-}$ is $(SiO_3)^{2-}$ (as $n = 3$). Thus, it is formed by sharing of two oxygen per Si atom and results in the formation of cyclic silicate.

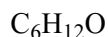


• = Silicon atom
○ = Oxygen atom

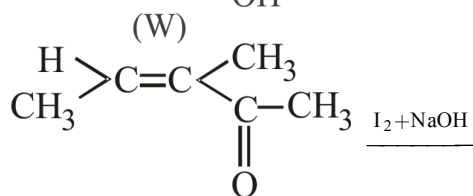
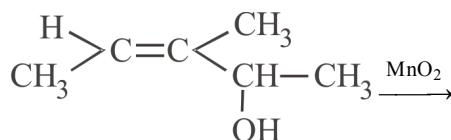
PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (D)

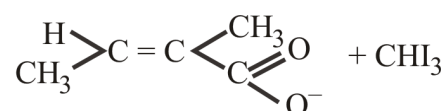


$$n = 1$$

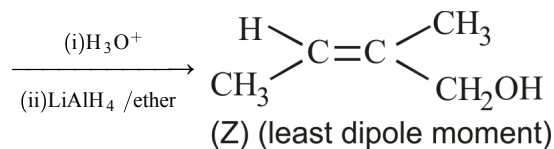


(X)

"Four isomers"

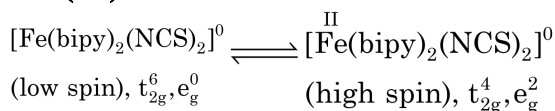


(Y) cannot decarboxylate on heating



(Z) (least dipole moment)

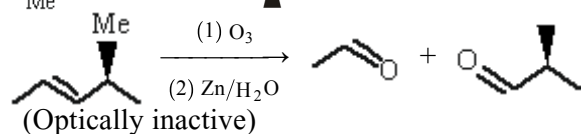
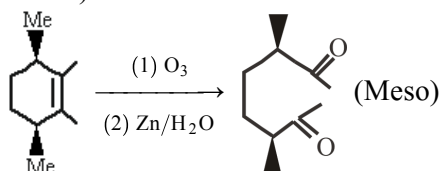
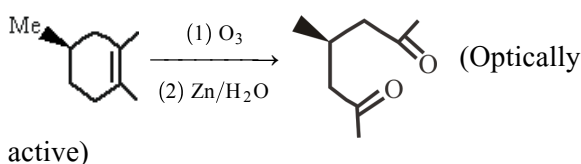
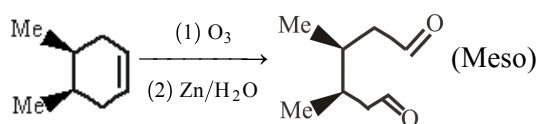
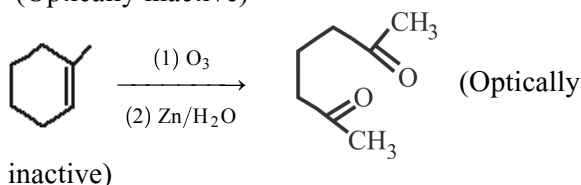
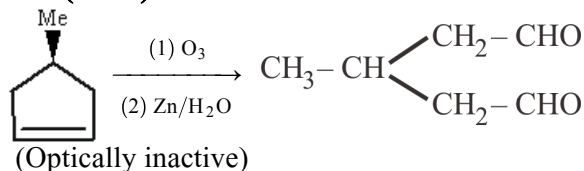
9. Ans (A)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (3.00)

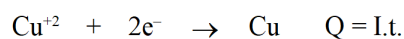


3. Ans (0.82)

$$n = M \times v = 2 \times 1$$

$$= 2 \text{ mole CuSO}_4$$

$$= 2 \text{ mole Cu}^{+2}$$



$$1 \text{ mol } 2F = 9.65 \times 2 \times 3600$$

$$1 \text{ mol } (2 \times 96500)C$$

$$x = \frac{(965 \times 72)C}{(2 \times 96500)C} = 965 \times 72 C$$

$$x = \frac{965 \times 72}{2 \times 96500} = \frac{36}{100} = 0.36 \text{ mol Cu}^{+2}$$

reduced to Cu

$$\text{Mole of Cu}^{+2} \text{ left in solution} = 2 - 0.36 = 1.64 \text{ mole}$$

$$[\text{Cu}^{+2}] = \frac{1.64}{2} = 0.82 \text{ mole/l}$$

4. Ans (21.48)

$$W_{\text{solution}} = 25 \text{ gm}$$

$$W_{\text{BaCl}_2} = 25 \times 0.12 = 3 \text{ gm}$$

$$W_{\text{BaCl}_2} \cdot 2\text{H}_2\text{O} = \frac{3 \times 244}{208} = 3.519 \text{ gm}$$

$$W_{\text{H}_2\text{O}} = 25 - 3.519$$

$$W_{\text{H}_2\text{O}} = 21.481 \text{ gm}$$

5. Ans (209.00)

$$\Delta u = 0.42 \text{ kJ} = 0.21 \text{ kJ/mol}$$

$$P = 2.7 \text{ bar} = 2.7 \times 10^5 \text{ pa}$$

$$\Delta V = V_{\text{(aragonite)}} - V_{\text{(Calcite)}}$$

$$= \left(\frac{100}{3.0} - \frac{100}{2.7} \right)$$

$$= -\frac{100}{27} \text{ cm}^3 = -\frac{100}{27} \times 10^{-6} \text{ m}^3$$

$$\Delta H = \Delta u + P \Delta V$$

$$= 0.21 \times 10^3 + 2.7 \times 10^5 \times \frac{100}{27} \times 10^{-6} \text{ m}^3$$

$$= 209 \text{ J}$$

6. Ans (20.00)

$$P = 6$$

$$Q = 4$$

(NCERT)

Correct b, e, f, h, i, j

Incorrect a, c, d, g

$$p^2 - q^2 = 36 - 16 = 20$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (C,D)

$$f^2(x) - [e^x + e^{-x}] f(x) + 1 = 0$$

$$(f(x) - e^x)(f(x) - e^{-x}) = 0$$

$$f(x) \text{ can be } e^x, e^{-x}, \begin{cases} e^x; & x \geq 0 \\ e^{-x}; & x < 0 \end{cases}, \begin{cases} e^{-x}; & x \geq 0 \\ e^x; & x < 0 \end{cases}$$

$$\int_{-1}^1 e^x dx = e - e^{-1}, \int_{-1}^0 e^{-x} dx + \int_0^1 e^x dx = 2e - 2$$

$$\int_{-1}^1 e^{-x} dx = e - e^{-1}, \int_{-1}^0 e^x dx + \int_0^1 e^{-x} dx = e - \frac{1}{e}$$

2. Ans (B,C)

Let $\theta = \cot^{-1} x_1$

$$\Rightarrow x_1 = \cot \theta; x_2 = \cot \theta + \sqrt{1 + \cot^2 \theta} = \cot \frac{\theta}{2}$$

$$x_3 = \cot \frac{\theta}{2^2} \text{ and so on } \dots$$

$$x_n = \cot \frac{\theta}{2^{n-1}} \text{ where } \theta = \frac{\pi}{6}$$

$$\alpha = \tan^{-1} y_1;$$

$$y_2 = \frac{\tan \alpha}{1 + \sqrt{1 + \tan^2 \alpha}} = \tan \frac{\alpha}{2}$$

$$\Rightarrow y_3 = \tan \frac{\alpha}{2^2} \text{ and so on } \dots$$

$$y_n = \tan \frac{\alpha}{2^{n-1}} \text{ where } \alpha = \frac{\pi}{3}$$

$$x_n y_n = \cot \frac{\pi}{3 \cdot 2^n} \tan \frac{\pi}{3 \cdot 2^{n-1}} = \cot \frac{\pi}{3 \cdot 2^n} \tan \frac{2\pi}{3 \cdot 2^n}$$

$$= \cot \frac{\pi}{3 \cdot 2^n} \left(\frac{2 \tan \frac{\pi}{3 \cdot 2^n}}{1 - \tan^2 \frac{\pi}{3 \cdot 2^n}} \right) = \frac{2}{1 - \tan^2 \frac{\pi}{3 \cdot 2^n}}$$

$$\text{At } n=1; x_1 y_1 = \frac{2}{1 - \tan^2 \frac{\pi}{6}} = 3$$

$$\text{As } n \uparrow; \frac{\pi}{3 \cdot 2^n} \downarrow; x_n y_n \text{ decreases}$$

As n goes to infinity; $x_n y_n$ tends to 2^+

$$k = \frac{x_n y_n}{\max} = 3, m = [x_n y_n] = 2$$

m^{2021} i.e. 2^{2021} is divided by 3, then remainder is 2

k^{2021} i.e. 3^{2021} is divided by 2, then remainder is 1.

3. Ans (A,B,C,D)

$$(A) P\left(\frac{A}{B^c}\right) = \frac{P(A \cap B^c)}{P(B^c)} = \frac{P(A) - P(A \cap B)}{1 - P(B)}$$

$$(B) P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$\Rightarrow P(A \cap B) \geq P(A) + P(B) - 1$$

$$(C) \text{ If } P\left(\frac{A}{B^c}\right) > P(A)$$

$$\frac{P(A \cap B^c)}{P(B^c)} > P(A)$$

$$\Rightarrow P(A) - P(A \cap B) > P(A) - P(A)P(B)$$

$$P(A \cap B) < P(A)P(B) \Rightarrow P\left(\frac{A}{B}\right) < P(A)$$

$$(D) P\left(\frac{A}{B^c}\right) + P\left(\frac{A^c}{B^c}\right) = 1$$

4. Ans (A,D)

$$A = \int_0^\infty \frac{\sin x}{x} dx$$

$$I = \int_0^\infty \frac{\sin ax \cos bx}{x} dx = \frac{1}{2} \left[\int_0^\infty \frac{\sin(a+b)x + \sin(a-b)x}{x} dx \right]$$

$$I = \frac{1}{2} \left[\int_0^\infty \frac{\sin(a+b)x}{x} dx + \int_0^\infty \frac{\sin(a-b)x}{x} dx \right]$$

$$I_1 = \int_0^\infty \frac{\sin(a+b)x}{x} dx.$$

$$\text{Let } (a+b)x = t \Rightarrow I_1 = \int_0^{(a+b)(\rightarrow \infty)} \frac{\sin t}{t} dt$$

$$I_2 = \int_0^\infty \frac{\sin(a-b)x}{x} dx.$$

$$\text{Let } (a-b)x = t \Rightarrow I_2 = \int_0^{(a-b)(\rightarrow \infty)} \frac{\sin t}{t} dt$$

$$I = \frac{1}{2} (I_1 + I_2)$$

If $a+b > 0$; then $I_1 = A$. If $a-b > 0$, then $I_2 = A$

$$\text{If } a-b < 0; \text{ then } I_2 = \int_0^{-\infty} \frac{\sin t}{t} dt = \int_0^\infty \frac{\sin t}{t} dt = A$$

$$\text{Now ; } a > b > 0 \Rightarrow I = \frac{1}{2} (A + A) = A$$

$$b > a > 0 \Rightarrow I = \frac{1}{2} (A - A) = 0$$

5. **Ans (A,C)**

$$xy - x - y = 3 \text{ implies } (x-1)(y-1) = 4$$

$$\text{Let } x-1 = X, y-1 = Y.$$

$$\text{Hence } XY = 4 \text{ has parametric form } \left(2t, \frac{2}{t}\right)$$

$$\text{In new co-ordinate system; } A(1, 4), B(4, 1), C(-2, -2)$$

$$A \left(t_1 = \frac{1}{2}\right), B(t_2 = 2), C(t_3 = -1)$$

$$\text{Orthocentre of } \triangle ABC \text{ is } \left(\frac{-2}{t_1 t_2 t_3}, -2t_1 t_2 t_3\right) \equiv (2, 2)$$

$$(x-1, y-1) \equiv (2, 2) \Rightarrow (x, y) \equiv (3, 3).$$

$$\text{Hence } \alpha = 3$$

[Note : you can also use the fact that orthocentre of $\triangle ABC$ lies on rectangular hyperbola only]

$$\text{No. of positive divisors of } 4\alpha^3 = 2^2 \cdot 3^3 \text{ is}$$

$$(1+2)(1+3) = 12$$

$$\text{No. of non-trivial positive divisors of } 5\alpha^6 = 5 \cdot 3^6 \text{ is}$$

$$(2 \times 7) - 1 - 1 = 12$$

6. **Ans (A,C,D)**

$S(n, k) \equiv$ No. of ways to partition n distinct items into k parts.

First consider one particular element a_1 forming one group and remaining $n-1$ elements forming $k-1$ partitions.

Second, consider $n-1$ elements forming k partitions and a_1 element going in one of ' k ' groups.

$$\text{Hence, } S(n, k) = S(n-1, k-1) + k S(n-1, k)$$

$$\text{No. of onto functions} = k! S(n, k)$$

$$= [k^n - {}^k C_1 (k-1)^n + {}^k C_2 (k-2)^n \dots \dots \dots]$$

$$S(5, 3) = \frac{1}{3!} [3^5 - {}^5 C_1 2^5 + {}^5 C_2 1^5] = 25$$

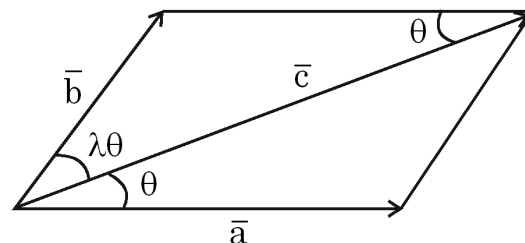
$$\text{No. of onto functions from 6 elements} = 3! S(6, 3) \text{ to 3 elements}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

$$\vec{c} = \vec{a} + \vec{b}, a > b$$



$$\frac{\sin \lambda \theta}{a} = \frac{\sin \theta}{b} = \frac{\sin(\pi - (\lambda + 1)\theta)}{c}$$

$$\text{For } \lambda = 2; \frac{\sin 2\theta}{a} = \frac{\sin \theta}{b}$$

$$\Rightarrow \cos \theta = \frac{a}{2b}$$

$$c = \frac{b \sin(\lambda + 1)\theta}{\sin \theta}$$

$$c = \frac{b \sin 3\theta}{\sin \theta} = b [3 - 4\sin^2 \theta]$$

$$c = \frac{a^2 - b^2}{b}$$

$$\text{For } \lambda = 3; \frac{\sin 3\theta}{a} = \frac{\sin \theta}{b} = \frac{\sin 4\theta}{c}$$

$$3 - 4\sin^2 \theta = \frac{a}{b} \text{ and}$$

$$c = b (2 \cos \theta) (2 \cos 2\theta)$$

$$c = 4b \cos \theta [1 - 2\sin^2 \theta]$$

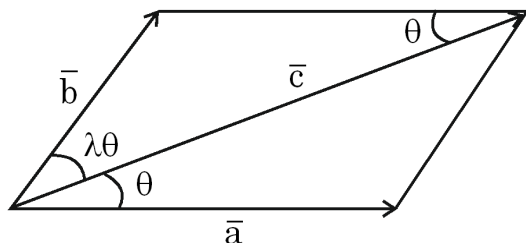
$$c = 4b \left(\frac{\sqrt{a+b}}{2\sqrt{b}} \right) \left[1 - \frac{3b-a}{2b} \right]$$

$$c = 4b \frac{\sqrt{a+b}}{2\sqrt{b}} \left(\frac{a-b}{2b} \right)$$

$$c = \frac{(a-b) \sqrt{a+b}}{\sqrt{b}}$$

8. Ans (A)

$$\bar{c} = \bar{a} + \bar{b}, a > b$$



$$\frac{\sin \lambda \theta}{a} = \frac{\sin \theta}{b} = \frac{\sin(\pi - (\lambda + 1)\theta)}{c}$$

$$\text{For } \lambda = 2; \frac{\sin 2\theta}{a} = \frac{\sin \theta}{b}$$

$$\Rightarrow \cos \theta = \frac{a}{2b}$$

$$c = \frac{b \sin(\lambda + 1)\theta}{\sin \theta}$$

$$c = \frac{b \sin 3\theta}{\sin \theta} = b [3 - 4\sin^2 \theta]$$

$$c = \frac{a^2 - b^2}{b}$$

$$\text{For } \lambda = 3; \frac{\sin 3\theta}{a} = \frac{\sin \theta}{b} = \frac{\sin 4\theta}{c}$$

$$3 - 4\sin^2 \theta = \frac{a}{b} \text{ and}$$

$$c = b(2 \cos \theta)(2 \cos 2\theta)$$

$$c = 4b \cos \theta [1 - 2\sin^2 \theta]$$

$$c = 4b \left(\frac{\sqrt{a+b}}{2\sqrt{b}} \right) \left[1 - \frac{3b-a}{2b} \right]$$

$$c = 4b \frac{\sqrt{a+b}}{2\sqrt{b}} \left(\frac{a-b}{2b} \right)$$

$$c = \frac{(a-b)\sqrt{a+b}}{\sqrt{b}}$$

9. Ans (B)

$$f(x) = \lim_{y \rightarrow \infty} \frac{1}{y \ln \left(\frac{\sin(x+1/y)}{\sin x} \right)}$$

$$= \lim_{t \rightarrow 0} \frac{t}{\ln(\sin(x+t)) - \ln(\sin x)}$$

$$f(x) = \lim_{t \rightarrow 0} \frac{1}{\frac{\cos(x+t)}{\sin(x+t)}} = \tan x$$

$$a = \lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} \frac{n}{(1 \cot^2 x + 2 \cot^2 x + \dots + (2n) \cot^2 x)^{\tan^2 x}}$$

$$a = \frac{1}{2} \lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} \frac{(2n)^{\cot^2 x \cdot \tan^2 x}}{(1 \cot^2 x + 2 \cot^2 x + \dots + (2n) \cot^2 x)^{\tan^2 x}}$$

$$a = \frac{1}{2} \lim_{n \rightarrow \infty} \lim_{x \rightarrow 0} \frac{1}{\left[\left(\frac{1}{2n} \right)^{\cot^2 x} + \left(\frac{2}{2n} \right)^{\cot^2 x} + \dots + \left(\frac{2n}{2n} \right)^{\cot^2 x} \right]^{\tan^2 x}}$$

$$\text{We have, } \lim_{x \rightarrow \infty} (p_1^x + p_2^x + \dots + p_k^x)^{1/x} = p_1$$

If p_1 is $\max^m$ of all p_i 's.

$$a = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{\left(\frac{2n}{2n} \right)} = \frac{1}{2}$$

$$b = \lim_{x \rightarrow 1} \left(\frac{4}{\pi} \tan^{-1} x \right)^{\frac{1}{x^2-1}}$$

$$b = e^{\lim_{x \rightarrow 1} \frac{\left(\frac{4}{\pi} \tan^{-1} x - 1 \right)}{x^2-1}}$$

$$\text{Let } \tan^{-1} x = t$$

$$b = e^{\lim_{t \rightarrow \frac{\pi}{4}} \frac{\left(\frac{4}{\pi} t - 1 \right)}{(\tan^2 t - 1)}}$$

$$b = e^{\lim_{t \rightarrow \frac{\pi}{4}} \frac{4\pi}{2 \tan t \sec^2 t}} = e^{1/\pi}$$

$$f(x) = ax \text{ has 5 roots in } (0, m\pi)$$

$$\Rightarrow \tan x = \frac{x}{2} \text{ has 5 roots in } (0, 6\pi)$$

$$m = 6$$

$$f(x) = bx \text{ has 6 roots in } (0, n\pi)$$

$$\Rightarrow \tan x = e^{1/\pi} x \text{ has 6 roots in } (0, 6\pi)$$

$$\text{Hence, } m + n = 12$$

10. Ans (A)

$$|\tan x| = \frac{1}{\pi} \text{ has 8 solutions in } (0, 4\pi) \text{ which is a multiple of 2.}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (0.40 to 0.43)

$$AB = \begin{bmatrix} 3x^2 \\ 1 \\ 6x \end{bmatrix} \begin{bmatrix} a & b & c \end{bmatrix} = \begin{bmatrix} 3x^2a & 3x^2b & 3x^2c \\ a & b & c \\ 6xa & 6xb & 6xc \end{bmatrix}$$

$$\text{tr}(AB) = \text{tr}(C)$$

$$\Rightarrow 3ax^2 + b + 6cx = (x + 2)^2 + 2x + 5x^2$$

$$\forall x \in \mathbb{R}$$

$$\Rightarrow a = 2, b = 4, c = 1$$

$$I = \int_0^{\infty} \frac{\ln x}{x^2 + 2x + 4} dx$$

$$\text{Substitute } x = 2t$$

$$I = \int_0^{\infty} \frac{\ln 2t}{4(t^2 + t + 1)} (2 dt) = \frac{1}{2} \int_0^{\infty} \frac{\ln 2 + \ln t}{t^2 + t + 1} dt$$

$$I = \frac{\ln 2}{2} \int_0^{\infty} \frac{dt}{t^2 + t + 1} + \frac{1}{2} \int_0^{\infty} \frac{\ln t}{t^2 + t + 1} dt$$

$$\text{Let, } I_1 = \int_0^{\infty} \frac{dt}{(t + \frac{1}{2})^2 + \frac{3}{4}}, I_2 = \int_0^{\infty} \frac{\ln t}{t^2 + t + 1} dt$$

$$I_1 = \frac{2}{\sqrt{3}} \tan^{-1} \frac{2t+1}{\sqrt{3}} \Big|_0^{\infty}$$

$$\Rightarrow I_1 = \frac{2}{\sqrt{3}} \left[\frac{\pi}{2} - \frac{\pi}{6} \right] = \frac{2\pi}{3\sqrt{3}}$$

$$I_2 = \int_0^{\infty} \frac{\ln t}{t^2 + t + 1} dt$$

$$\text{Let } t = \frac{1}{u}$$

$$I_2 = \int_{\infty}^0 \frac{-\ln u \left(-\frac{1}{u^2} du \right)}{\frac{u^2+u+1}{u^2}}$$

$$= - \int_0^{\infty} \frac{\ln u du}{u^2 + u + 1} = -I_2 \Rightarrow I_2 = 0$$

$$I = \left(\frac{\ln 2}{2} \right) \frac{2\pi}{3\sqrt{3}}$$

$$I = \frac{\pi \ln 2}{3\sqrt{3}} = 0.419$$

2. Ans (0.10)

$$S_n = \sum_{k=1}^n \frac{(-1)^{k-1} {}^nC_k}{k}$$

$$= \frac{{}^nC_1}{1} - \frac{{}^nC_2}{2} + \frac{{}^nC_3}{3} - \dots + (-1)^{n-1} \frac{{}^nC_n}{n}$$

$$(1-x)^n = {}^nC_0 - {}^nC_1x + {}^nC_2x^2 - {}^nC_3x^3 + \dots + (-1)^n {}^nC_nx^n$$

$$\int_0^1 \frac{1 - (1-x)^n}{x} dx$$

$$= \int_0^1 \left[{}^nC_1 - {}^nC_2x + {}^nC_3x^2 - \dots + (-1)^{n-1} {}^nC_nx^{n-1} \right] dx$$

$$S_n = \frac{{}^nC_1}{1} - \frac{{}^nC_2}{2} + \frac{{}^nC_3}{3} - \dots + (-1)^{n-1} \frac{{}^nC_n}{n}$$

$$= \int_0^1 \frac{1-x^n}{1-x} dx = \int_0^1 (1+x+x^2+\dots+x^{n-1}) dx$$

$$= 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$$

$$S_n - S_{n-1}$$

$$= \left[1 + \frac{1}{2} + \dots + \frac{1}{n} \right] - \left[1 + \frac{1}{2} + \dots + \frac{1}{n-1} \right] = \frac{1}{n}$$

$$S_{10} - S_9 = 0.1$$

3. Ans (5.00)

$$f(x+1) \leq f(x) + 1$$

$$\Rightarrow f(x+2) \leq f(x+1) + 1 \leq f(x) + 1 + 1$$

$$\text{i.e. } f(x+2) \leq f(x) + 2$$

$$\Rightarrow f(x+3) \leq f(x+1) + 2 \leq f(x) + 1 + 2$$

$$\Rightarrow f(x+3) \leq f(x) + 3$$

$$\text{Continuing same way } f(x+5) \leq f(x) + 5$$

$$\text{But } f(x+5) \geq f(x) + 5$$

$$\text{Hence, } f(x+5) = f(x) + 5$$

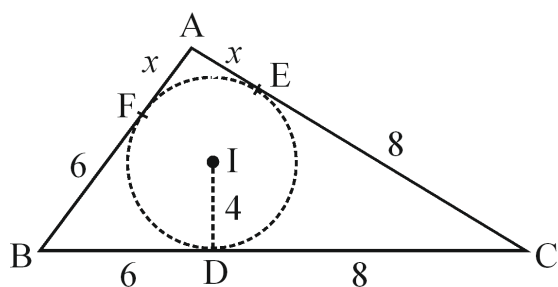
$$\text{All equalities must hold simultaneously, } f(x+1)$$

$$= f(x) + 1$$

$$\text{or } f(x) = x$$

$$g(x) = f(x) + 5 - x \Rightarrow g(2013) = 5$$

4. Ans (1.72 to 1.74)



$$s = x + 6 + 8$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\Delta = \sqrt{(x+14)(x)(8)(6)}$$

$$\sqrt{8 \times 6x(x+14)} = 4(x+14)$$

$$\Rightarrow x = 7 \Rightarrow \Delta = 84$$

$$\sqrt{\Delta - 81} = \sqrt{3}$$

5. Ans (1.50)

6 boys and 6 girls are to be arranged such that no. of girls $\geq$ no. of boys at any position from ticketing window.

$$\text{Number of ways} = C_6 \times 6! \times 6!$$

where C_n denotes catalan

Number of index n which can be given as

$$C_n = \frac{{}^{2n}C_n}{n+1}$$

$$\text{Hence, no. of ways} = C_6 \times (720)^2$$

$$= \frac{{}^{12}C_6}{7} \times (720)^2 = 132 \times 720^2$$

$$\Rightarrow \lambda = 132$$

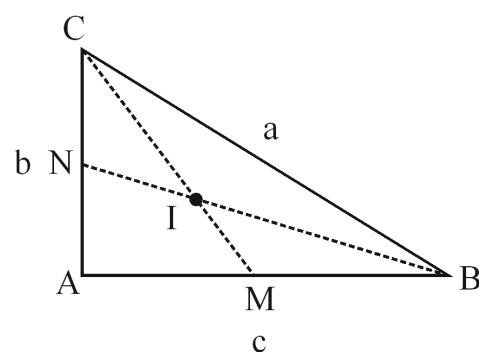
6. Ans (1.57)

$$\frac{a}{R} = 2, \frac{b}{R} = \frac{3}{2};$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

$$\Rightarrow \sin A = 1, \sin B = \frac{3}{4}$$

$$\Rightarrow \cos B = \frac{\sqrt{7}}{4}$$



ΔABC is right angled at A

$$\frac{BN}{CM} = \frac{c \sec B/2}{b \sec C/2}$$

$$= \frac{2R \sin C \cos C/2}{2R \sin B \cos B/2} = \frac{\cos B \cos C/2}{\sin B \cos B/2}$$

$$\frac{BN}{CM} = \frac{\sqrt{7}}{3} \frac{\cos(\frac{\pi}{4} - \frac{B}{2})}{\cos \frac{B}{2}} = \frac{\sqrt{7}}{3\sqrt{2}} \left[1 + \tan \frac{B}{2} \right]$$

$$\sin B = \frac{2 \tan \frac{B}{2}}{1 + \tan^2 \frac{B}{2}} = \frac{3}{4}$$

$$\Rightarrow 3t^2 - 8t + 3 = 0$$

$$\Rightarrow \tan \frac{B}{2} = \frac{4 - \sqrt{7}}{3} \left[\text{as } \frac{B}{2} < 45^\circ \right]$$

$$\text{Hence, } \frac{BN}{CM} = \frac{7}{9\sqrt{2}} (\sqrt{7} - 1)$$

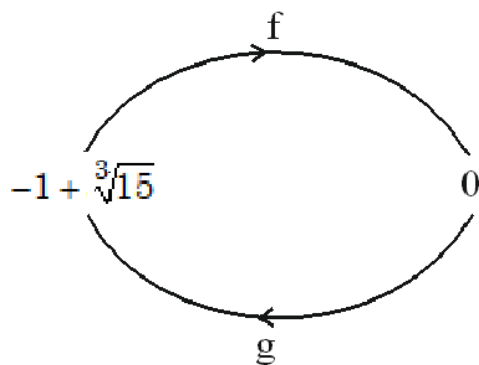
7. Ans (0.00)

$$x^3 + y^3 + 3(x^2 + y^2) + 3(x + y) = 14$$

$$(x + 1)^3 + (y + 1)^3 = 16$$

$$y = -1 + [16 - (x + 1)^3]^{1/3}$$

$$f(-1 + \sqrt[3]{15}) = 0$$



Because of symmetry $f^{-1}(x) = g(x) = f(x)$ only

$$h(x) = \frac{d}{dx} [f(x + g(x)) \cdot g(x + f(x))]$$

$$= \frac{d}{dx} [f(x + f(x)) \cdot f(x + f(x))]$$

$$h(x) = 2f(x + f(x)) f'(x + f(x)) [1 + f'(x)]$$

$$h(-1 + \sqrt[3]{15})$$

$$= 2f(-1 + \sqrt[3]{15} + 0) f'(-1 + \sqrt[3]{15} + 0) [1 + f'(-1 + \sqrt[3]{15})]$$

$$= 2(0) f'(-1 + \sqrt[3]{15}) [1 + f'(-1 + \sqrt[3]{15})] = 0$$

8. Ans (0.29 or 0.30)

$$f_n(x) = \min \left\{ \frac{x^n}{n!}, \frac{(1-x)^n}{n!} \right\}; 0 \leq x \leq 1$$

$$f_n(x) = \begin{cases} \frac{x^n}{n!} & ; 0 \leq x < \frac{1}{2} \\ \frac{(1-x)^n}{n!} & ; \frac{1}{2} \leq x \leq 1 \end{cases}$$

$$I_n = \int_0^1 f_n(x) dx = \int_0^{1/2} \frac{x^n}{n!} dx + \int_{1/2}^1 \frac{(1-x)^n}{n!} dx$$

$$= 2 \int_0^{1/2} \frac{x^n}{n!} dx$$

$$I_n = 2 \left[\frac{1}{2^{n+1}} \frac{1}{(n+1)!} \right]$$

$$\sum_{n=1}^{\infty} I_n = 2 \left[\frac{1}{2^2 2!} + \frac{1}{2^3 3!} + \frac{1}{2^4 4!} + \dots \right]$$

$$e^x - 1 - x = \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sum_{n=1}^{\infty} I_n = 2 \left[e^{1/2} - 1 - \frac{1}{2} \right] = 2\sqrt{e} - 3$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

FULL SYLLABUS

31-07-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,B	B,D	A,C,D	B,C,D	A,C		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	D	C	A	A				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	2.00	1.25	2.57	1.00	2.00	40.00	90.00	55.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,C,D	A,C,D	A,B,D	A,B,C,D	B,C,D	A,B		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	B	D	A	C				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	5.00	8.00	39.25	3.00	1.14	10.76	5.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6		
	A.	A,B,C,D	A,B,C	B,D	B,C,D	B,D	B,D		
SECTION-I (ii)	Q.	7	8	9	10				
	A.	A	C	C	B				
SECTION-II	Q.	1	2	3	4	5	6	7	8
	A.	5.00	2.71 to 2.72	1.41	180.00	40.00	-3.14	87.00	1.66 to 1.67

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. **Ans (A,C,D)**

For the equilibrium of the satellite,

$$\frac{Gmm'}{r^2} + m'\omega^2(R-r) = \frac{GMm'}{(R-r)^2}$$

$$\text{Also, } \omega^2 = \frac{GM}{R^3}$$

$$\therefore Gm \frac{R^2}{r^2} + GM \left(1 - \frac{r}{R}\right) = GM \left(1 + \frac{2r}{R}\right)$$

[After multiplying both sides by R^2 and using the binomial approximation].

$$\therefore \frac{m}{3M} = \frac{r^3}{R^3}$$

$$\therefore r = R \sqrt[3]{\frac{m}{3M}}$$

$$= 1.5 \times 10^{11} \sqrt[3]{\frac{6 \times 10^{24}}{2 \times 10^{30} \times 3}}$$

$$= 1.5 \times 10^9 \text{ m}$$

$$T' = \frac{2\pi}{\sqrt{GM}} \times r^{3/2}$$

$$= \frac{2\pi}{\sqrt{GM}} \times R^{3/2} \times \sqrt{\frac{M}{m}} \times \left(\frac{r}{R}\right)^{3/2}$$

$$= 1 \text{ year} \times \sqrt{\frac{1}{3} \times 10^6} \times \frac{1}{1000} = 7 \text{ months}$$

2. Ans (A,B)

At the earth's distance, the radiation pressure $2S/c$ pushes the craft away from the sun, and gravity, mg pulls the craft towards the sun. Both these factors are inversely proportional to r^2 . (r = distance from sun)

$$F_{\text{net}} = ma$$

$$\therefore \frac{2SA}{c} \times \frac{R^2}{r^2} - mg \times \frac{R^2}{r^2} = m \frac{d^2r}{dt^2}$$

$$\therefore \frac{d^2r}{dt^2} = 0 \text{ if } A = \frac{mgc}{2S}$$

$$= \frac{140 \times 6 \times 10^{-3} \times 3 \times 10^8}{2 \times 1400}$$

$$= 9 \times 10^4 \text{ m}^2$$

If the area is 50% more than this

$$0.5 mg \times \frac{R^2}{r^2} = m \times v \frac{dv}{dr}$$

Integrating from $r = R$ to $r = \infty$,

$$v = \sqrt{gR} = 3 \times 10^4 \text{ m/s}$$

3. Ans (B,D)

There is a slight deviation due to the rotation of the earth.

$$F_x = ma_x$$

$$\therefore m \frac{d^2x}{dt^2} = m \times 2v_{\text{rot}} \omega$$

$$= 2 mgt \omega$$

$$\therefore x = g \frac{t^3}{3} \omega$$

$$= \frac{g}{3} \times \frac{2\pi}{T} \times \left(\frac{2h}{g} \right)^{3/2}$$

$$= \frac{10}{3} \times \frac{2\pi}{24 \times 3600} \left(\frac{2 \times 720}{10} \right)^{3/2}$$

$$= \frac{10}{3} \times \frac{2\pi}{2 \times 12 \times 12 \times 300} \times 12^3$$

$$= \frac{2\pi}{15} = 0.42 \text{ m}$$

$\vec{v}$ is downward.

$\vec{\omega}$ is from south to north.

Hence, $\vec{v} \times \vec{\omega}$ is eastward.

5. Ans (B,C,D)

$$J = \sigma E$$

$$B = \frac{\mu_0 J_x}{2}$$

$$B \times 2\pi r = \mu_0 \frac{I r^2}{R^2}$$

$$\therefore B = \frac{\mu_0 I r}{2\pi R^2} \quad (r < R)$$

$$u = \int \frac{B^2}{2\mu_0} dv$$

$$= \int \frac{1}{2\mu_0} \times \frac{\mu_0^2 I^2 r^2}{4\pi^2 R^4} \times 2\pi r dr \ell$$

$$= \frac{\mu_0 I^2 \ell}{16\pi}$$

6. Ans (A,C)

$$\text{Rate of heat lost} = [K_1(2hl + 2hb + lb) + K_2lb] \times \Delta T \times 2$$

$$= 1.16 \times 2 \Delta T$$

$$= \sigma \Delta T$$

$$\text{Hence the heat lost is } H_1 = 1.16 \times 2 \times 20 \times 86400 \times 4$$

$$= 1.6 \times 10^7 \text{ J.}$$

If it is allowed to cool:

$$mc \frac{dT}{dt} = -\sigma (T - T_s)$$

$$\text{Hence } T - T_s = (T_0 - T_s) \times e^{(-\sigma t/mc)}$$

Hence

$$T = 5 + 20 \times e^{(-1.16 \times 2 \times 4 \times 86400 / 20 \times 1 \times 2 \times 1000 \times 4000)}$$

$$= 5 + 20 \times e^{-0.5} = 5 + 20 \times 0.6 = 17^\circ\text{C}$$

When it is heated from 17°C back to 25°C

$$H_2 - 0.10H_2 = mc(25 - 17)$$

$$H_2 = 1.4 \times 10^7 \text{ J}$$

$H_1 > H_2$; Hence option 2 is better.

PART-1 : PHYSICS

SECTION-I (ii)

7. **Ans (D)**

For $e \rightarrow i$

$$45^\circ > \theta_c$$

$$\sin 45^\circ > \sin \theta_c$$

$$\frac{1}{\sqrt{2}} > \frac{\mu_2}{\mu_1}$$

$$\mu_1 > \sqrt{2} \mu_2$$

For $e \rightarrow f$

angle of refraction is lesser than angle of incidence,

so $\mu_2 > \mu_1$ and then $\mu_2 > \mu_3$

For $e \rightarrow g$, $\mu_1 = \mu_2$

for $e \rightarrow h$, $\mu_2 < \mu_1 < \sqrt{2} \mu_2$ and $\mu_2 > \mu_3$

8. **Ans (C)**

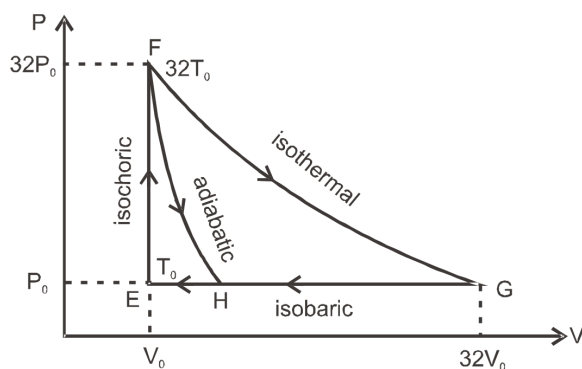
(P) In α decay mass number decreases by 4 and atomic number decreases by 2.

(Q) In β^+ decay mass number remains unchanged while atomic number decreases by 1.

(R) In Fission, parent nucleus breaks into almost two equal fragments.

(S) In proton emission both mass number and atomic number decreases by 1.

9. **Ans (A)**



In $F \rightarrow G$ work done in isothermal process is

$$nRT \ln \left(\frac{V_f}{V_i} \right) = 32 P_0 V_0 \ln \left(\frac{32 V_0}{V_0} \right) \\ = 32 P_0 V_0 \ln 2^5 = 160 P_0 V_0 \ln 2$$

In $G \rightarrow E$,

$$\Delta W = P_0 \Delta V = P_0 (31 V_0) = 31 P_0 V_0$$

In $G \rightarrow H$ work done is less than $31 P_0 V_0$ i.e.,

$$24 P_0 V_0$$

In $F \rightarrow H$ work done is $36 P_0 V_0$

10. **Ans (A)**

$$x = \frac{t^3}{3} - 2t^2 + 3 + 5$$

$$v = t^2 - 4t + 3 = t^2 - 3t - t + 3 = (t-1)(t-3)$$

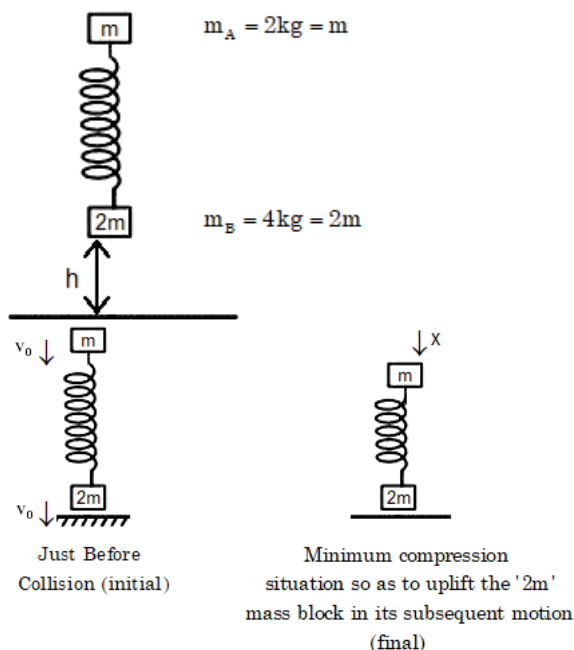
$$a = 2t - 4 = 2(t-2)$$

t=1	t=2	t=3
$V > 0$	$V < 0$	$V < 0$
$a < 0$	$a < 0$	$a > 0$

PART-1 : PHYSICS

SECTION-II

1. Ans (2.00)



To lift up the '2m' i.e. 'B' block extension in the spring must be $\frac{2mg}{k}$.

Now applying conservation of energy B/w initial and final situation

$$0 - \frac{1}{2}mv_0^2 = -mgx - \frac{1}{2}kx^2$$

$$\frac{1}{2}mv_0^2 = mgx + \frac{1}{2}kx^2$$

$$\text{put } x = \frac{2mg}{k}$$

$$h = \frac{4mg}{k} = \frac{4 \times 2 \times 10}{40} = 2$$

3. Ans (2.57)

Time period = Time taken to travel to and fro between C and D plus half time period with left spring + half time period with right spring

$$t = t_1 + t_2 + t_3$$

$$= 2 \times \frac{60}{120} + \frac{1}{2}2\pi\sqrt{\frac{m}{k_1}} + \frac{1}{2}2\pi\sqrt{\frac{m}{k_2}}$$

$$= \frac{120}{120} + \pi\sqrt{\frac{0.2}{3.2}} + \pi\sqrt{\frac{0.2}{3.2}}$$

$$= 1 + \frac{\pi}{4} + \frac{\pi}{4} = 1 + \frac{\pi}{2} = 1 + \frac{3.14}{2}$$

$$= 1 + 1.57$$

$$= 2.57$$

4. Ans (1.00)

Speed decreasing to zero uniformly. Let a be a positive constant (acceleration) then

$$v = a(T - t)$$

$$s = \int_0^T v dt = \left[aTt - \frac{1}{2}at^2 \right]_0^T$$

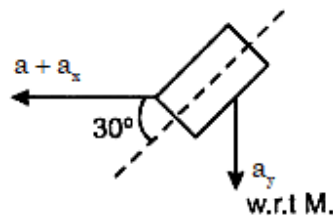
$$= aT^2 - \frac{aT^2}{2} = \frac{1}{2}aT^2$$

$$a = \frac{2s}{T^2}, v = \frac{2S}{T^2}(T - t)$$

$$v = \frac{2S}{T^2}(T - t) = 1 \text{ m/s}$$

5. Ans (2.00)

a_x and a_y component of acceleration of m and a acceleration of M.



$$\frac{a_y}{a + a_x} = \frac{1}{\sqrt{3}}$$

$$a + a_x = \sqrt{3}a_y = 3\sqrt{3}$$

$$mg - N \cos 30^\circ = ma_y$$

$$N \sin 30^\circ = ma_x$$

$$a_y = \left| \frac{d^2h}{dt^2} \right| = 3$$

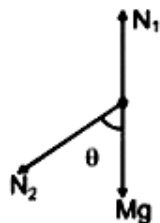
on solving

$$a = \frac{2}{\sqrt{3}}$$

$$x = 2$$

6. Ans (40.00)

Acceleration of the system $a = \frac{F}{2M}$



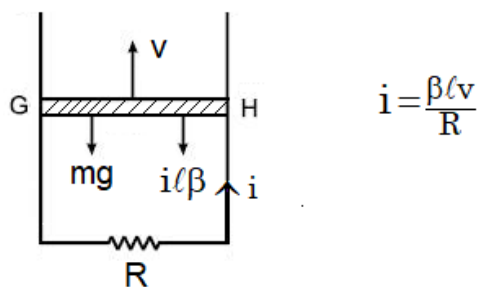
$$N_2 \sin \theta = Ma = \frac{F}{2}$$

$$N_1 = Mg + N_2 \cos \theta$$

$$= Mg + \frac{F}{2} \cot \theta$$

$$= 20 + \frac{30}{2} \times \frac{4}{3} = 40 \text{ N}$$

7. Ans (90.00)



$$(F_{\text{net}}) GH = -m \frac{dv}{dt} = mg + i\ell\beta$$

$$= -\frac{dv}{dt} = g + \frac{i\ell\beta}{m}$$

$$\Rightarrow -\frac{dv}{g + \frac{i\ell\beta}{m}} = dt$$

$$\Rightarrow -\int_{1=\mu}^0 \frac{dv}{g + \frac{\beta^2 \ell^2 v}{Rm}} = \int_0^t dt$$

$$\text{Let } \frac{\beta^2 \ell^2}{mR} = \alpha$$

$$= \frac{(3)^2 (0.1 \text{ m})^2}{(10^{-3} \text{ kg}) (1\pi)} = 90$$

$$-\int_1^0 \frac{dv}{g + 90v} = \int_0^t dt$$

$$-\frac{\log_e}{90} (g + 90v) \Big|_1^0 = t$$

$$\frac{\log_e}{90} \left(\frac{10 + 90}{10} \right) = t$$

$$\frac{\log_e (10)}{90} = t$$

$$x = 90$$

8. Ans (55.00)

Photon of wavelength just more than 4400 Å will be corresponding to transition 4 → 2 of an e⁻

$$\text{i.e. } \frac{1}{\lambda_{4-2}} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) \approx 4860 \text{ Å}$$

Now a photon incident on metal plate of work function = 2eV is of energy (E).

$$E = 13.6 \left(\frac{1}{2^2} - \frac{1}{4^2} \right) \text{ eV} = 2.55 \text{ eV}$$

$$eV_s = E_p - \phi = 2.55 - 2.00 \text{ eV,}$$

$$V_s = 0.55 \text{ volt} = 55 \times 10^{-2} \text{ volt}$$

PART-2 : CHEMISTRY

SECTION-I (i)

1. Ans (A,C,D)

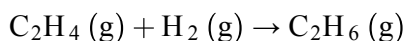
$$\frac{t_{50\%}}{t_{75\%}} = \frac{\log_{10} \left(\frac{100}{100-50} \right)}{\log_{10} \left(\frac{100}{100-75} \right)}$$

$$= \frac{\log_{10} 2}{\log_{10} 4} = \frac{\log_{10} 2}{2 \cdot \log_{10} 2}$$

$$\frac{t_{50\%}}{t_{75\%}} = \frac{1}{2} \quad t_{75\%} = 2t_{50\%}$$

$$K = A \cdot e^{-E_a/RT}$$

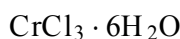
K increases exponentially with increase in temperature.



$$\text{rate} = K [\text{C}_2\text{H}_4]^1$$

order = 1.

2. Ans (A,C,D)



$$\pi = iCRT \quad i = 1 + \alpha (n - 1)$$

$$98.4 = i \times 2 \times 0.082 \times 300$$

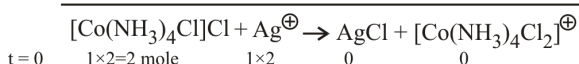
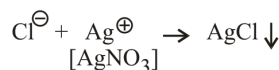
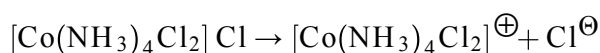
$$i = 2$$

$$1 + \alpha (n - 1) = 2$$

$$1 + n - 1 = 2 \quad (\alpha = 1)$$

$$\boxed{n = 2}$$

CoCl₃ · 4NH₃ can be written as



$$t = 0 \quad \begin{array}{ccccccc} & 1 \times 2 = 2 \text{ mole} & & 1 \times 2 = 2 \text{ mole} & & 0 & & 0 \\ & & & = 2 \text{ mole} & & & & \end{array}$$

$$\text{Weight of AgCl} = 2 \times 143.5 = 287 \text{ g}$$

$$\text{Final Solution} = 2 \text{ mole } [\text{Co}(\text{NH}_3)_4\text{Cl}_2] \text{NO}_3$$

$$\text{Total volume} = 1 + 2 = 3 \text{ litre}$$

$$C = \frac{2}{3}$$

$$\pi = iCRT \quad \left| \begin{array}{l} i = 1 + \alpha (n - 1) \end{array} \right.$$

$$= 2 \times \frac{2}{3} \times 0.082 \times 300 \quad \left| \begin{array}{l} = 1 + \alpha (2 - 1) \end{array} \right.$$

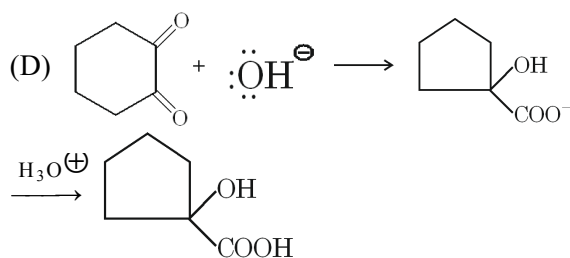
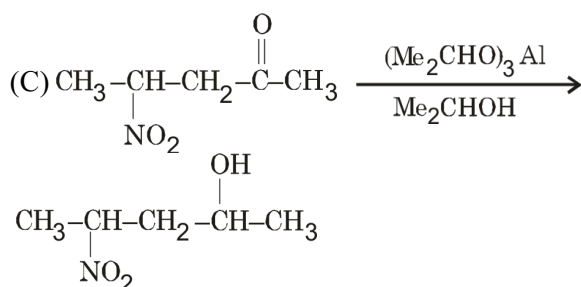
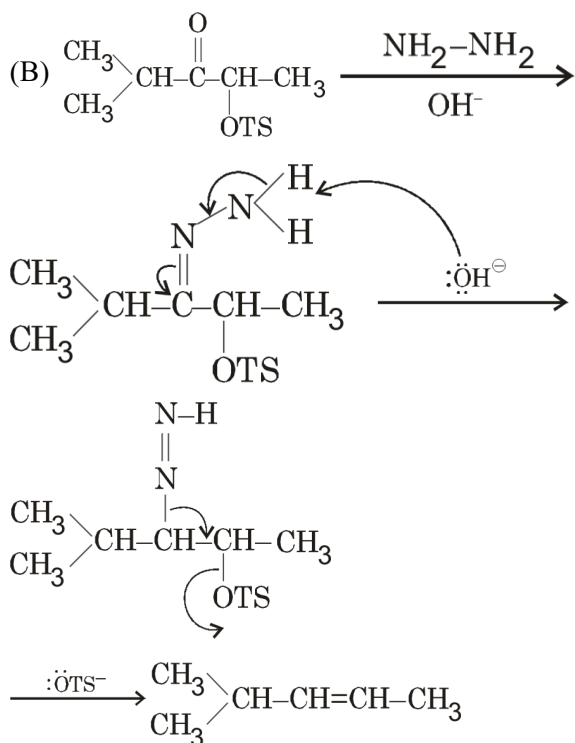
$$\quad \left| \begin{array}{l} = 1 + \alpha \end{array} \right.$$

$$\pi = 32.8 \text{ atm} \quad \left| \begin{array}{l} = 1 + 1 \end{array} \right.$$

$$\quad \left| \begin{array}{l} i = 2 \end{array} \right.$$

3. **Ans (A,B,D)**

(A) Acid catalysed halogenation



4. **Ans (A,B,C,D)**

Non Stoichiometric NaCl has F-centre due to metal excess defect.

5. **Ans (B,C,D)**

$\text{Ag}_3(\text{PO}_4)$ yellow

AlPO_4 , $\text{Ba}_3(\text{PO}_4)_3$ and MgNH_4PO_4 are white

PART-2 : CHEMISTRY

SECTION-I (ii)

7. **Ans (B)**

$$\text{MA}_2\text{BCDE} \quad \text{GI} = 9 \quad \text{SI} = 15 \quad (\text{OAI}) = 12$$

$$\text{MA}_3\text{BCD} \quad \text{GI} = 4 \quad \text{SI} = 5 \quad (\text{OAI}) = 2$$

$$\text{M}(\text{AB})_3 \quad \text{GI} = 2 \quad \text{SI} = 4 \quad (\text{OAI}) = 4$$

$$\text{MA}_2\text{B}_2\text{C}_2 \quad \text{GI} = 5 \quad \text{SI} = 6 \quad (\text{OAI}) = 2$$

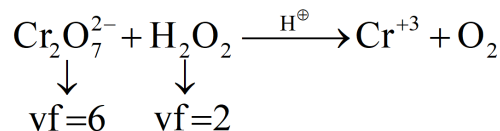
PART-2 : CHEMISTRY

SECTION-II

1. **Ans (5.00)**

SiO_2 , P_4O_{10} , P_4O_8 , CrO_3 , Mn_2O_7

2. **Ans (5.00)**



No. of meq. of $\text{K}_2\text{Cr}_2\text{O}_7$ = no. of meq. of H_2O_2

$$40 \times \frac{1}{24} \times 6 = 11.35 \times M \times 2$$

$$M = \frac{5}{11.35}$$

$$\text{Volume Strength of } \text{H}_2\text{O}_2 = \frac{5}{11.35} \times 11.35 = 5$$

3. **Ans (8.00)**

Truncated tetrahedron has 4 hexagonal faces and 4 triangular faces.

4. **Ans (39.25)**

For real gas.

$$\left(P + \frac{an^2}{v^2} \right) (v - nb) = nRT$$

$$P = \left(\frac{nRT}{v - nb} \right) - \frac{n^2a}{v^2}$$

$$= \left(\frac{0.5 \times 0.083 \times 475}{0.5 - 0.5 \times 0.05} \right) - \frac{(0.5)^2 \times 2.25}{(0.5)^2}$$

$$= 39.25 \text{ bar}$$

5. Ans (3.00)

$$\Lambda_m^\infty(\text{AgCl}) = \Lambda_m^\infty(\text{AgNO}_3) + \Lambda_m^\infty(\text{NaCl}) - \Lambda_m^\infty(\text{NaNO}_3)$$

$$= 116.5 + 110.3 - 105.2$$

$$= 121.6 \text{ S cm}^2 \text{ mol}^{-1}$$

For sparingly soluble salt,

$$\Lambda_m = \Lambda_m^\infty = 121.6 \text{ S} \cdot \text{cm}^2 \text{ mol}^{-1}$$

$$K_{\text{AgCl}} = (3.592 - 1.16) \times 10^{-6}$$

$$K_{(\text{AgCl})} = 2.432 \times 10^{-6} \Omega^{-1} \text{ cm}^{-1}$$

$$\Lambda_m = \frac{K \times 1000}{S}$$

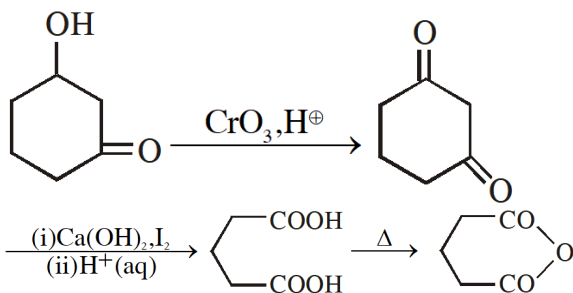
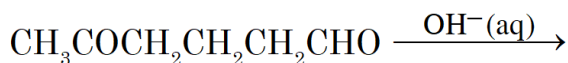
$$121.6 = \frac{2.432 \times 10^{-6} \times 1000}{S}$$

$$S = 2 \times 10^{-5} \text{ M}$$

$$= 2 \times 10^{-5} \times 143.5 \text{ g/L}$$

$$= 2.87 \times 10^{-3} \text{ g/L}$$

6. Ans (1.14)



$$M = 114 \text{ gm}$$

7. Ans (10.76)

$$pI = \frac{pK_{a2} + pK_{a3}}{2} = \frac{9.04 + 12.48}{2} = 10.76$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,B,C,D)

$$\begin{aligned} & \frac{1}{\sqrt{2} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{8}} + \frac{1}{\sqrt{8} + \sqrt{11}} + \dots n \text{ terms} \\ &= \frac{\sqrt{5} - \sqrt{2}}{3} + \frac{\sqrt{8} - \sqrt{5}}{3} + \frac{\sqrt{11} - \sqrt{8}}{3} + \\ & \dots + \frac{\sqrt{5 + (n-1)3} - \sqrt{2 + (n-1)3}}{3} \\ &= \frac{\sqrt{3n+2} - \sqrt{2}}{3} \\ &= \frac{3n+2-2}{3(\sqrt{3n+2} + \sqrt{2})} \\ &= \frac{n}{\sqrt{3n+2} + \sqrt{2}} \\ &= \frac{n}{\sqrt{2+3n} + \sqrt{2}} < \frac{n}{\sqrt{3n}} < n \end{aligned}$$

2. Ans (A,B,C)

Any point on $xy = 2$ is $P \left(\sqrt{2}t, \frac{\sqrt{2}}{t} \right)$

Normal at P is $xt - \frac{y}{t} = \sqrt{2} \left(t^2 - \frac{1}{t^2} \right)$.

Put $x = 3, y = 4$ to get a polynomial equation of degree 4

$$\begin{aligned} 3t - \frac{4}{t} &= \sqrt{2} \left(t^2 - \frac{1}{t^2} \right) \\ \Rightarrow \sqrt{2}t^4 - 3t^3 + 4t - \sqrt{2} &= 0. \end{aligned}$$

Roots are t_1, t_2, t_3, t_4 .

$$\sum t_1 = \frac{3}{\sqrt{2}}, \sum \frac{1}{t_1} = \frac{4}{\sqrt{2}} \quad \text{and} \quad \prod t_1 = -1.$$

3. Ans (B,D)

For real roots $4a^2 - 4b(a-1) > 0$

$$\Rightarrow a^2 - ba + b > 0$$

Now for above inequality to hold for all values of $a, b^2 - 4b < 0$ or $0 < b < 4$.

4. **Ans (B,C,D)**

If $n = m$ then equation becomes $z^n = (\bar{z})^n$ and it has infinite solution because any $z \in \mathbb{R}$ will satisfy it.

If $n \neq m$, let $n > m$, then $z^n = (\bar{z})^m$

$$\Rightarrow |z|^n = |z|^m \Rightarrow |z|^m(|z|^{(n-m)} - 1) = 0$$

$$\Rightarrow |z| = 0 \text{ or } |z| = 1, |z| = 0$$

$$\Rightarrow z = 0 + 0i \text{ } |z| = 1$$

$$\Rightarrow z^{n+m} = |z|^{2m}$$

$$\Rightarrow z^{m+n} = 1 \text{ hence}$$

number of solution is $m + n + 1$

where number of non-zero solution is equal to $m + n$ and one solution is $z = 0$

5. **Ans (B,D)**

$$F(x) = \int_x^{x^2} \frac{dt}{\ln t}$$

$$\Rightarrow F'(x) = \frac{2x}{\ln(x^2)} - \frac{1}{\ln x} = \frac{x-1}{\ln x}$$

Thus for $x > 1$, $F(x)$ is strictly increasing.

6. **Ans (B,D)**

$$\text{Curve is } (y-2)^2 = 2x^3 - 4 \quad \dots(1)$$

$$\frac{dy}{dx} = \frac{3x^2}{y-2}$$

$$\text{Eqn. of tangent at } (x_1, y_1) \text{ is } y - y_1 = \frac{3x_1^2}{y_1 - 2}(x - x_1)$$

As it passes through $(1, 2)$

$$\Rightarrow 2 - y_1 = \frac{3x_1^2}{y_1 - 2}(1 - x_1)$$

$$\Rightarrow (y_1 - 2)^2 = 3x_1^2(1 - x_1)$$

$$\Rightarrow 2x_1^3 - 4 = 3x_1^3 - 3x_1^2$$

$$[\text{as } (y_1 - 2)^2 = 2x_1^3 - 4]$$

$$\Rightarrow x_1^3 - 3x_1^2 + 4 = 0$$

$$\Rightarrow (x_1 + 1)(x_1 - 2)^2 = 0$$

$$\Rightarrow x_1 = -1 \text{ or } 2$$

From eqn. (1) it is clear that $x_1 \neq -1$

$$\text{Thus } x_1 = 2 \Rightarrow y_1 = 2 \pm 2\sqrt{3}$$

Points of contact are

$$(2, 2 + 2\sqrt{3}) \text{ and } (2, 2 - 2\sqrt{3})$$

$\therefore$ sum of x-coordinates of points of contact is 4

$\therefore$ sum of y-coordinates of points of contact is 4.

PART-3 : MATHEMATICS

SECTION-I (ii)

7. **Ans (A)**

$$(P) a_1 = \int_0^{\pi/2} dx = \frac{\pi}{2}, a_2 = \pi$$

$$\text{Also, } 2a_{n+1} - a_{n+2} - a_n$$

$$= \int_0^{\pi/2} \frac{2 \cos(2n+2)x - \cos(2n+4)x - \cos(2nx)}{1 - \cos 2x} dx$$

$$= \int_0^{\pi/2} \frac{2 \cos(2n+2)x - 2 \cos(2n+2)x \cos 2x}{1 - \cos 2x} dx$$

$$= \int_0^{\pi/2} 2 \cos(2n+2)x dx$$

$$= \frac{2}{2n+2} (\sin(2n+2)x)_0^{\pi/2} = 0$$

$\therefore a_1, a_2, a_3, \dots$ are in A.P.

$$(Q) I_{n+1} + I_{n+3}$$

$$= \int_0^{\pi/4} \tan^{n+1} x \sec^2 x dx$$

$$= \frac{1}{n+2} \tan^{n+2} x \Big|_0^{\pi/4} = \frac{1}{n+2}$$

$$\therefore a_n = n + 2$$

$\Rightarrow a_1, a_2, \dots$ are in A.P.

$$(R) I_n - I_{n-1} = \int_0^{\pi/2} \frac{\sin^2(nx) - \sin^2(n-1)x}{\sin x} dx$$

$$= \int_0^{\pi/2} \frac{\sin(2n-1)x \sin x}{\sin x} dx$$

$$= -\frac{1}{2n-1} \cos(2n-1)x \Big|_0^{\pi/2} = \frac{1}{2n-1}$$

$\Rightarrow a_1, a_2, a_3, \dots$ are in H.P.

$$(S) b_n = \int_0^{\pi/2} \frac{1 - \cos 2nx}{1 - \cos 2x} dx = \frac{n\pi}{2} \quad (\text{See (a)})$$

$$a_n = \frac{\pi}{2} \quad \forall n$$

$\therefore a_1, a_2, a_3, \dots$ are in A.P., G.P. and H.P.

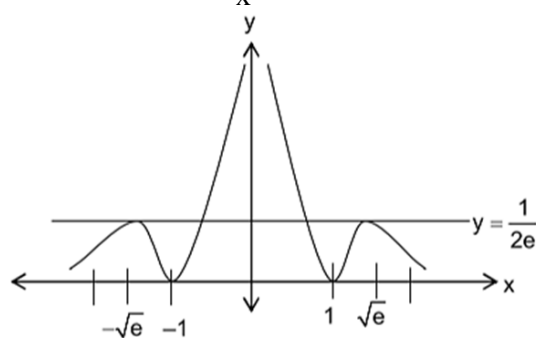
Also $a_1, a_2, a_3, \dots$ is a constant sequence.

8. Ans (C)

$$f(x) = |\ln|x|| - kx^2 = 0$$

$$\Rightarrow k = \frac{|\ln|x||}{x^2}$$

Consider $g(x) = \frac{|\ln|x||}{x^2}$ as shown



9. Ans (C)

$$(P) x = r \cos \theta, y = r \sin \theta$$

$$\Rightarrow x^2 + y^2 = r^2 \text{ and } \tan \theta = \frac{y}{x}$$

$$\Rightarrow x dx + y dy = r dr \text{ and}$$

$$\frac{x dy - y dx}{x^2} = \sec^2 \theta d\theta$$

$$\Rightarrow x dy - y dx = x^2 \sec^2 \theta d\theta$$

$$= r^2 \cos^2 \theta \sec^2 \theta d\theta$$

$$\Rightarrow x dy - y dx = r^2 d\theta$$

So given differential equation is

$$\frac{r dr}{r^2 d\theta} = \sqrt{\frac{a^2 - r^2}{r^2}}$$

$$\Rightarrow \frac{dr}{\sqrt{a^2 - r^2}} = d\theta$$

$$\Rightarrow \sin^{-1} \left(\frac{r}{a} \right) = \theta + c$$

$$(Q) \rightarrow 4$$

$$(R) \rightarrow 3$$

(S) Diff. w.r.t. to x

$$\Rightarrow xy - \int_0^x y(t) dt = (x+1)xy + \int_0^x ty(t) dt$$

again d w.r.t. to x

$$y = x^2 y' + 2xy + xy$$

$$\Rightarrow \frac{(1-3x)}{x^2} dx = \frac{dy}{y}$$

$$\Rightarrow y = \frac{c}{x^3} e^{-1/x}$$

10. Ans (B)

As the given line lies in the plane $3 - 15 - 2\alpha = 0$

and $2 + 3 + 2\alpha + \beta = 0 \Rightarrow \alpha = -6$ and $\beta = 7$

$$(P) \text{ number of terms} = {}^{\beta+r-1}C_{r-1} = {}^9C_2$$

$$(Q) \sum_{i=0}^{n-r} {}^{n-i}C_r = {}^nC_r + {}^{n-1}C_r + {}^{n-2}C_r + \dots + {}^rC_r$$

$$= {}^{n+1}C_{r+1} \Rightarrow \text{For } n = 13, r = 2 \Rightarrow {}^{14}C_3$$

$$(R) x_1 x_2 x_3 = 42 = 2 \times 3 \times 7$$

Number of positive integral solutions

$$= {}^3C_2 \times {}^3C_2 \times {}^3C_2 = 27$$

one positive and two negative

$$= {}^3C_1 \times 27 = 81$$

$$\text{Total integral solutions} = 27 + 81 = 108$$

(S) Perpendicular distance from the point

$$= \left| \frac{1 + 3 - \alpha + \beta}{\sqrt{1 + 9 + 36}} \right| = \frac{17}{\sqrt{46}}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5.00)

Apply the following row operations :

$$R_1 \rightarrow R_1 - R_2$$

$$R_2 \rightarrow R_2 - R_3$$

$$R_3 \rightarrow R_3 - R_1$$

$$D = \begin{vmatrix} a & -b & 0 & 0 \\ 0 & b & -c & 0 \\ 0 & 0 & c & -d \\ 1 & 1 & 1 & 1 \end{vmatrix}$$

$$= abcd + (abc + abd + acd + bcd)$$

$$= 3 + 2 = 5$$

2. Ans (2.71 to 2.72)

$$\ln(g(x)) = \lim_{r \rightarrow 0} \frac{\ln((x+1)^{r+1} - x^{r+1})}{r}$$

$$= \lim_{r \rightarrow 0} \frac{(x+1)^{r+1} \ln(x+1) - x^{r+1} \ln x}{(x+1)^{r+1} - x^{r+1}}$$

$$= (x+1) \ln(x+1) - x \ln x = \ln \left(\frac{(x+1)^{x+1}}{x^x} \right)$$

$$\Rightarrow g(x) = \frac{(x+1)^{x+1}}{x^x}$$

$$\Rightarrow \frac{g(x)}{x} = \left(1 + \frac{1}{x} \right)^x \left(1 + \frac{1}{x} \right)$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{g(x)}{x} = (e)(1) = e$$

3. Ans (1.41)

Equation of curve is $\frac{\left(\frac{x+y}{\sqrt{2}}\right)^2}{2} + \frac{\left(\frac{x-y-2}{\sqrt{2}}\right)^2}{8} = 1$.

This is an ellipse with $a = 2\sqrt{2}$, $b = \sqrt{2}$ and center $C(1, -1)$. L is a normal to the ellipse. Greatest possible distance of a normal to an ellipse from its centre is equal to $a - b = \sqrt{2}$

4. Ans (180.00)

$$\text{Let } S_n = \sum_{r=(n+10)^2}^{(n+100)^2} \frac{1}{\sqrt{r}}.$$

Note that S_n is a strictly decreasing function of n , and has $180n + 9901$ terms.

Thus,

$$\frac{180n+9901}{\sqrt{n^2+200n+10000}} < S_n < \frac{180n+9901}{\sqrt{n^2+20n+100}}.$$

By the sandwich theorem, $\lim_{n \rightarrow \infty} S_n = 180$

5. Ans (40.00)

$$\begin{aligned} & 16k^4 + 24k^2 + 25 \\ &= (4k^2 + 5)^2 - 16k^2 \\ &= (4k^2 - 4k + 5)(4k^2 + 4k + 5) \end{aligned}$$

$$\begin{aligned} \text{Thus, } t_k &= \frac{k}{16k^4 + 24k^2 + 25} \\ &= \frac{1}{8} \left\{ \frac{1}{(4k^2 - 4k + 5)} - \frac{1}{(4k^2 + 4k + 5)} \right\} \end{aligned}$$

$$S = \frac{1}{8} \left\{ \frac{1}{5} - 0 \right\} = \frac{1}{40}$$

6. Ans (-3.14)

Let $f(x) = x^3 + bx^2 + cx + 1$

$$f(0) = 1 > 0, f(-1) = b - c < 0$$

So, $\alpha \in (-1, 0)$

So, $2\tan^{-1}(\operatorname{cosec} \alpha) + \tan^{-1}(2 \sin \alpha \sec^2 \alpha)$

$$= 2 \tan^{-1} \left(\frac{1}{\sin \alpha} \right) + \tan^{-1} \left(\frac{2 \sin \alpha}{1 - \sin^2 \alpha} \right)$$

$$= 2 \left[\tan^{-1} \left(\frac{1}{\sin \alpha} \right) + \tan^{-1}(\sin \alpha) \right]$$

$$= 2 \left(-\frac{\pi}{2} \right) = -\pi \text{ (as } \sin \alpha < 0)$$

7. Ans (87.00)

$$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$$

$$= 3 \sum |\vec{a}|^2 - 2 \sum \vec{a} \cdot \vec{b}$$

$$= 58 - 2 \sum \vec{a} \cdot \vec{b} \quad \dots(1)$$

$$|\vec{a} + \vec{b} + \vec{c}|^2$$

$$= \sum |\vec{a}|^2 + 2 \sum \vec{a} \cdot \vec{b}$$

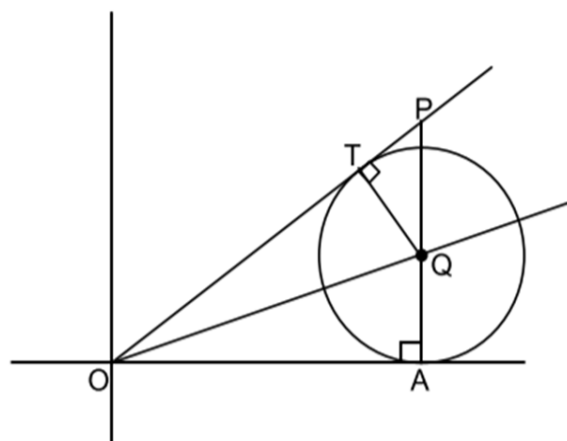
$$= 29 + 2 \sum \vec{a} \cdot \vec{b} \quad \dots(2)$$

From (1) and (2), we get

$$|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$$

$$= 87 - |\vec{a} + \vec{b} + \vec{c}|^2 \leq 87$$

8. Ans (1.66 to 1.67)



Given $QT = QA = 1$

Let $PQ = x$, then $PT = \sqrt{x^2 - 1}$

Then $\triangle TQR$ and $\triangle APO$ are similar triangles

$$\text{Then } OT = OA = \frac{x+1}{\sqrt{x^2-1}}$$

$$\Rightarrow 1 + x + \frac{2(x+1)}{\sqrt{x^2-1}} + \sqrt{x^2-1} = 8$$

$$\Rightarrow x = \frac{5}{3}$$